

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset $\text{\LaTeX}$ solutions.

1.a

$\text{Rain} \rightarrow \text{WetGrass}$ and $\text{Sprinkler} \rightarrow \text{WetGrass}$. There is no edge between Rain and Sprinkler, since neither one causes the other.

1.b

The joint is the product of each variable's local conditional probability given its parents:

$$\mathbb{P}(\text{Rain} = r, \text{Sprinkler} = s, \text{WetGrass} = w) = p(r) p(s) p(w \mid r, s).$$

1.c

Sum the joint over both values of Rain and both values of Sprinkler:

$$\mathbb{P}(\text{WetGrass} = w) = \sum_r \sum_s p(r) p(s) p(w \mid r, s).$$

1.d

The grass is wet with probability 0.9 when it rained and the sprinklers were on, so it stays dry with probability $1 - 0.9 = 0.1$:

$$\begin{aligned} \mathbb{P}(\text{Rain} = 1, \text{Sprinkler} = 1, \text{WetGrass} = 0) &= p(\text{Rain} = 1) p(\text{Sprinkler} = 1) p(\text{WetGrass} = 0 \mid \text{Rain} = 1, \text{Sprinkler} = 1) \\ &= 0.2 \times 0.5 \times 0.1 = 0.01. \end{aligned}$$

1.e

- (a) True.
- (b) False.
- (c) True.

2.d

One run of `forward_sampling` with `mutation_rate = 0.1` and `genome_length = 10`:

Species	Genome
Thomas bayus	TTCCATTCCC
Humblus studentus	TGCCATTCCCT
Aryamus bayus	TTCCATTCCC
Kenius bayus	TTCCATTCCC

2.g

Estimates of $\mathbb{P}(\text{ThomasBayus} = \text{AAAC} \mid \text{KeniusBayus} = \text{AAAA})$, from `test_gibbs_vs_rejection` with `mutation_rate = 0.1`, `genome_length = 4`, seed 0:

Method	Steps/Iterations	Estimated Posterior
Rejection sampling	100	0.0000
Rejection sampling	10,000	0.0667
Gibbs sampling	100	0.0900
Gibbs sampling	10,000	0.0332
Exact	—	0.0335

3.a

$Y \rightarrow A_1$, $Y \rightarrow A_2$ and $Y \rightarrow A_3$. The true label Y is what each annotator's claimed label depends on, and the annotators do not influence one another, so there are no edges between them.

3.b

A perfect annotator always gives the true label.

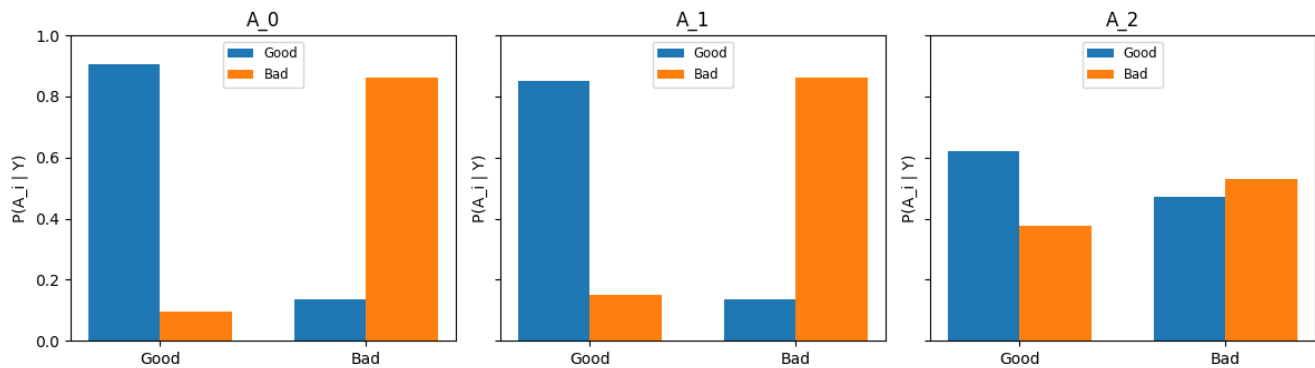
Y	$p(A_1 = \text{good} \mid Y)$	$p(A_1 = \text{bad} \mid Y)$
good	1	0
bad	0	1

3.c

A random annotator ignores the true label, so both rows are the same.

Y	$p(A_1 = \text{good} \mid Y)$	$p(A_1 = \text{bad} \mid Y)$
good	0.5	0.5
bad	0.5	0.5

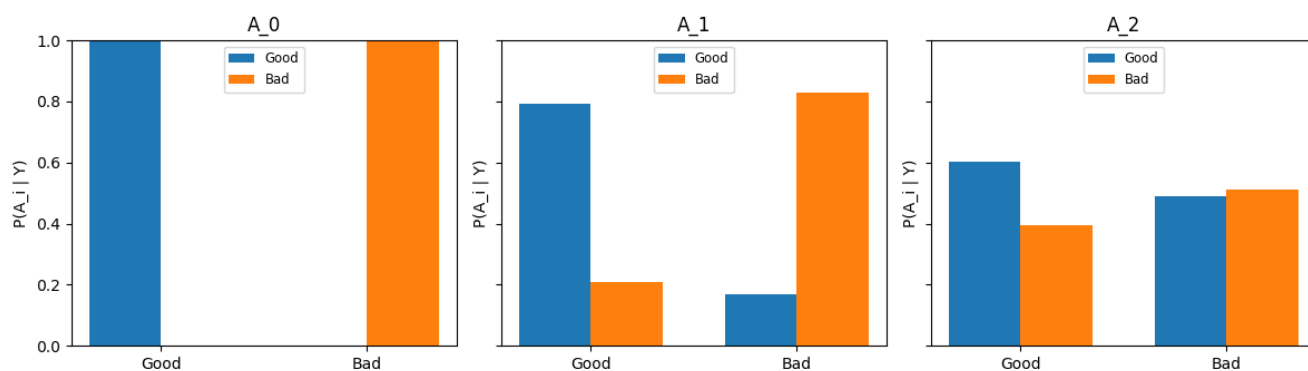
3.g



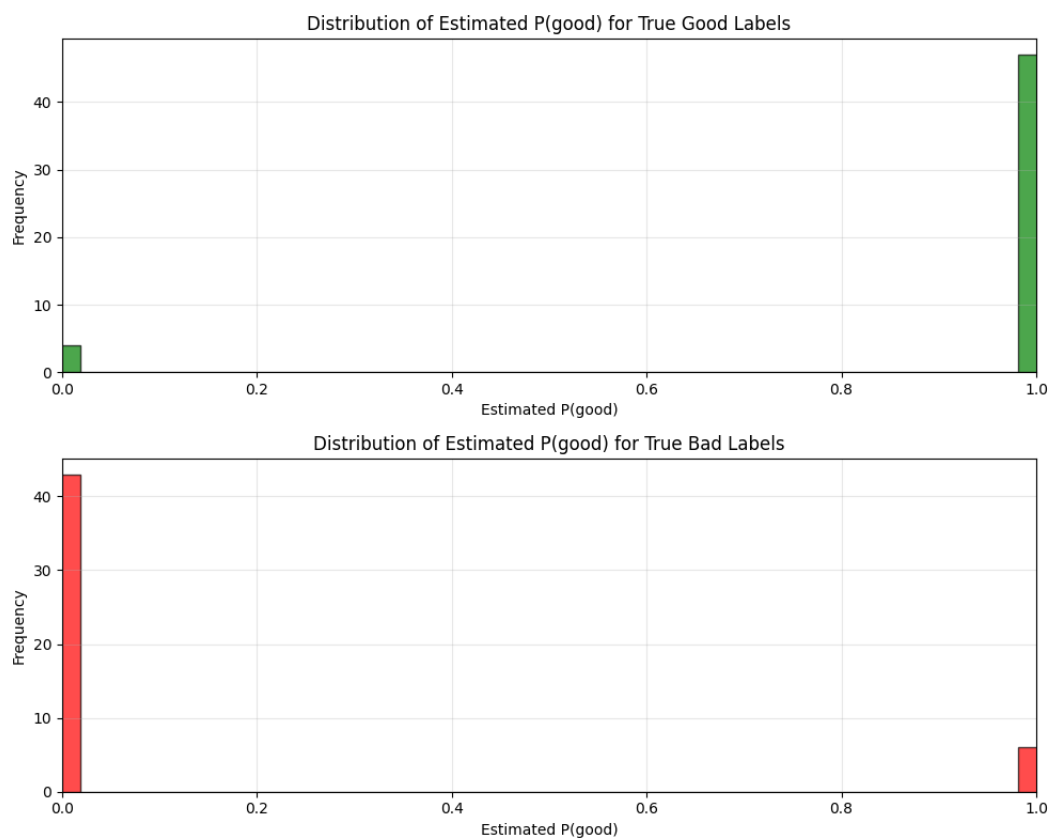
A_2 is the least trustworthy. Its answers are close to a coin flip whichever the true label is: when the conversation is good it says good about 62% of the time, and when it is bad it only says bad about 53% of the time, so its table looks much more like the random annotator from 3c than the perfect one from 3b. A_0 and A_1 both get the label right around 85 to 90% of the time either way.

4.d

Annotator tables after 100 iterations of EM on the unlabelled annotations:



Estimated $P(\text{good})$ for each data point, split by its true label:



4.e

Only partly. About 90 of the 100 data points end up on the right side: most of the truly good ones get an estimated $P(\text{good})$ near 1 and most of the truly bad ones near 0, with a handful on the wrong side in each. But the annotator plot shows EM has turned A_0 into a perfect annotator, so what it really learned is A_0's labels, treated as the truth. That works reasonably well here because A_0 was the most accurate annotator, right about 90% of the time when trained with the true labels in 3g, and the data points EM gets wrong are the ones A_0 got wrong.

5.a

5.b

5.c

5.d

6.a
6.b